



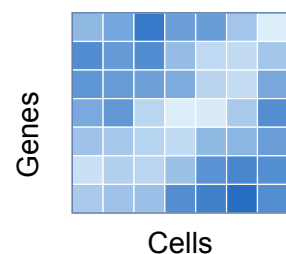
ATLAS

Advanced Trajectory Learning
from multi-omics At
Single-cell resolution

A. INPUT

MuData object

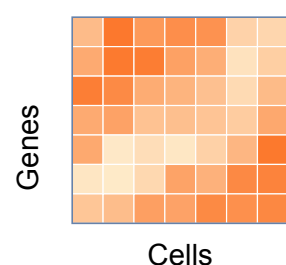
scRNA-seq



scATAC-seq

or

Gene activity

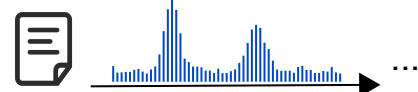


Cell metadata

(e.g., cell type)

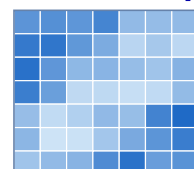


Fragment file (optional)

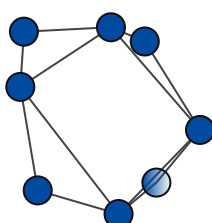


B. CONSTRUCT WNN GRAPH

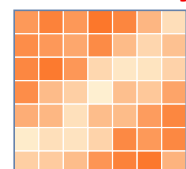
scRNA-seq



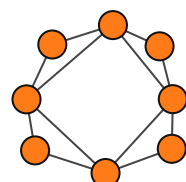
RNA graph



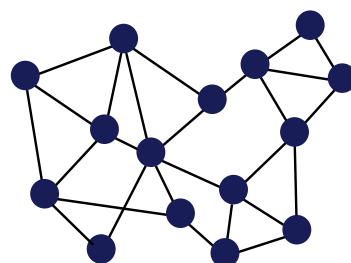
Gene activity



Gene activity graph



Weighted Nearest Neighbors (WNN) graph

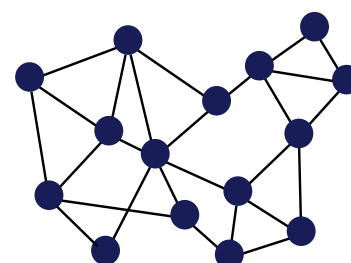


joint cell-state space capturing
complementary transcriptomic and
accessibility information

C. TRAJECTORY INFERENCE

Input

WNN graph



Palantir

CellRank

Fate probabilities



Entropy



Pseudotime



Terminal states



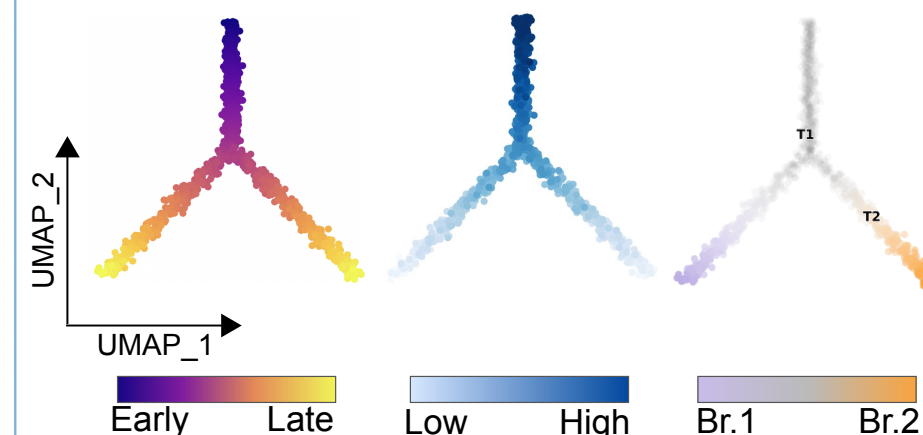
D. OUTPUTS AND VISUALIZATIONS

Cell-state dynamics on embeddings

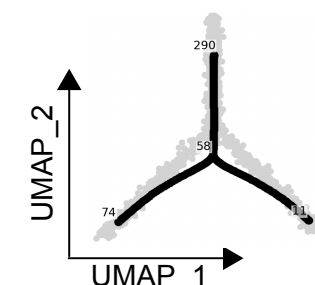
Pseudotime

Entropy

Fate probabilities



Differentiation topology



Multi-omic regulation along pseudotime

TF expression (RNA)

Gene activity

